

NAKEDI TUMELO PETA

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PROFESSIONAL SUMMARY

Motivated and technically skilled Computer Systems Engineering student with hands-on experience in cybersecurity, full-stack development, and cloud security. Demonstrated ability to design and deploy end-to-end security systems, conduct SOC operations, and apply machine learning techniques for threat detection. Holds certifications from Deloitte, EC-Council, IT Varsity, and SANCS, with CompTIA Security+ and MWR CyberSecurity currently in progress.

CORE COMPETENCIES

Cybersecurity & SOC Operations

- SOC Operations | Log Analysis | Incident Detection | Threat Hunting
- Penetration Testing | Vulnerability Assessment | Web Security
- Azure Cloud Security | Honeypot Design | GeoIP Log Enrichment
- Machine Learning for Security: Isolation Forest | Anomaly Detection | Feature Extraction
- Kali Linux | Social Engineer Toolkit (SET) | File Integrity Monitoring (Wazuh/Syscheck)
- Intrusion Detection | Network Security | Computer Networks

Software & Systems Development

- Python | JavaScript | C++ | SQL | HTML | CSS
- Flask | OpenCV | Firebase | ESP32 | Arduino | Embarcadero C++
- Docker | Linux (Ubuntu) | Azure | GitHub | VS Code | Cursor
- Full-Stack Web Development | UI/UX Design | REST APIs | Client-Server Architecture
- Embedded Systems | IoT Hardware Integration | PVC Enclosure Fabrication

Professional Skills

- Analytical Problem-Solving | Attention to Detail | Self-Directed Learning
- Technical Documentation | Dashboard Reporting | Cross-Functional Collaboration

PROJECT EXPERIENCE

AI-Powered Facial Recognition Access Control System | *Independent Project* | 2025

Repository: github.com/naksMann/facial-recognition-access-control

- Architected and deployed a production-ready smart security system using Python Flask, OpenCV face recognition, Firebase Realtime Database, and ESP32 microcontroller, delivering automated access control suitable for residential and commercial environments.
- Engineered a full-stack responsive web dashboard featuring live video monitoring, role-based user management, and real-time email alerts for unauthorised access events, reducing manual security monitoring effort.
- Designed and fabricated a professional PVC hardware enclosure housing all electronic components, ensuring the system was deployment-ready and physically robust.
- Achieved facial recognition accuracy of 95% in controlled light conditions.

Azure Honeypot & Attack Heatmap | *Cybersecurity Monitoring Project* | 2025

Repository: github.com/naksMann/honeypot-threat-analysis

- Deployed and configured a Microsoft Azure-based honeypot environment designed to attract, capture, and analyse malicious traffic from global threat actors.
- Integrated Azure Monitor and Log Analytics to collect attacker activity logs, enriched data with GeoIP geolocation, and generated an interactive heatmap visualising global attacker origins.

- Identified and documented key attack vectors and regional threat patterns, producing actionable security intelligence from real-world attacker data.
- Captured real-world malicious traffic within 24 hours of deployment, identifying attackers originating from 3 distinct countries, validating the effectiveness of the honeypot configuration and GeoIP enrichment pipeline.

SOC Insider Threat Detection Lab | *Wazuh + Machine Learning + Docker* | 2025

Repository: github.com/naksMann/soc-insider-threat-detection

- Architected a Docker-based Security Operations Centre (SOC) simulation lab using Wazuh agents configured to monitor both normal user behaviour and simulated insider threat activity.
- Implemented File Integrity Monitoring (Syscheck) on sensitive directories and generated controlled file events to create a labelled dataset for anomaly detection modelling.
- Applied the Isolation Forest machine learning algorithm to behavioural features extracted from Wazuh logs, successfully detecting anomalous file size deviations and modification patterns indicative of data exfiltration.
- Visualised threat alerts and detection outputs via Wazuh dashboards, demonstrating end-to-end insider threat detection capability in a realistic SOC environment.

EDUCATION

Diploma in Computer Systems Engineering | *Tshwane University of Technology (TUT)* | Jan 2022 – Oct 2026 (Expected)

- Campus: Soshanguve South
- Key areas: Embedded systems, computer networks, software engineering, electronics

National Senior Certificate | *Olievenhoutbosch Secondary School* | Jan 2017 – Dec 2021

- Location: Centurion, Gauteng

CERTIFICATIONS & TRAINING

- **EC-Council:** Data Science in Cybersecurity
- **Deloitte:** Cyber Security Job Simulation
- **South African Netherlands Cyber Security School (SANCS):** Attendance & Participation Certificate
- **IT Varsity:** Full Stack Development Certificate
- **WeThinkCode:** Prompt Engineering Certificate
- **CompTIA Security+:** In Progress
- **MWR CyberSecurity Certification:** In Progress